A robot that acts to know

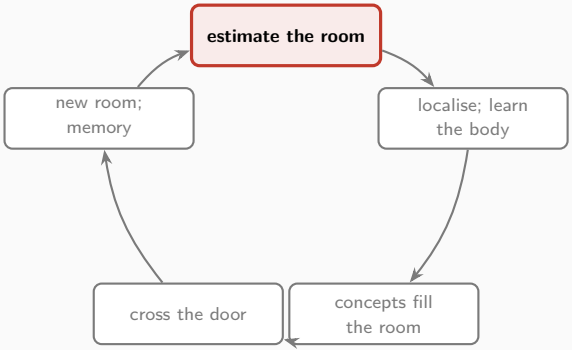
Day 2 — Estimating the room, and finding oneself in it

Course built on the doctoral thesis of Noé Zapata

Tuesday 1 December

Montevideo, 30 November – 4 December 2026

Where we are



Yesterday the robot knew its pose, and we told it which points were the wall's.

Today it knows neither:

- which points belong to which wall: **association**;
- how many walls there are: **structure**;
- where it is in the room it is building: **localisation**.

And every one of these is coupled to the others.

Two regimes

The room is either a variable or a constant

Acquisition: the room is a variable

- Walls (α_w, d_w) are unknowns, and so is their *number*.
- Two poses that see one wall are coupled through it, so the graph has **cycles**.
- This is SLAM-like: poses and walls are solved together, and the solve *terminates*.

Operating: the room is a constant

- The polygon is given; only the pose is estimated.
- The factors are unary, so the graph is a **chain**: the information matrix is banded and the solve is linear in the window length.
- The published covariance is over the pose alone.

acquisition $\xrightarrow{\text{the polygon, exactly Manhattan}}$ operating

operating $\xrightarrow{\text{the calibrated body}}$ acquisition

source: layout_acquisition.tex l.131–174

Why split them? Because they change at different rates

- A room changes when someone moves a wall: over years. A pose changes on every cycle.
- Estimating both jointly at frame rate, for ever, tracks a near-constant at great expense. It is the same category error as promoting a calibration constant to a state.
- So: **estimate jointly while the room is in question, then stop**, and hand over a *settled* belief.

Why split them? Because they change at different rates

- A room changes when someone moves a wall: over years. A pose changes on every cycle.
- Estimating both jointly at frame rate, for ever, tracks a near-constant at great expense. It is the same category error as promoting a calibration constant to a state.
- So: **estimate jointly while the room is in question, then stop**, and hand over a *settled* belief.

The price

The operating regime cannot detect that the polygon it was given is wrong. The graph has no loop, so there is no loop closure: a systematic map error is absorbed into the poses while the reported covariance stays confident.

You will measure this price in the lab.

source: layout_acquisition.tex l.178–208

Acquiring the room

What is estimated

quantity	what it is	why it is carried
$\mathcal{O}$	cyclic list of wall identifiers	it *is* the topology; changing it changes the dimension of everything else
α_w, d_w	each wall's line, $n(\alpha_w)^\top p = d_w$	what the LiDAR constrains directly
Λ_w	each wall's 2×2 information	values the next look (Day 5)
$[s_{\min}, s_{\max}]_w$	observed extent	a wall is a segment, not a line
$e_w[i]$	existence log-odds per 25 cm	a wall can be too *long*, not just misplaced
k_w, θ_0	Manhattan class, room direction	prefer right angles without forbidding others
X_t	five-pose window	walls are variables, so the poses must be solved with them

The state is both **discrete** ($\mathcal{O}$, the number of walls) and **continuous** (α, d). No number of Gauss–Newton steps changes a discrete variable, so a change of shape must be *priced*.

source: layout_acquisition.tex l.271–327

Whose frame? A gauge, and why it matters

- With no map there is no world frame. The robot's frame at $t = 0$ becomes the map frame.
- That is a **gauge**: a convention no measurement could ever fix, because the data would be identical in any other frame.
- The room's dominant direction θ_0 is found by a **quadrupled-angle vote**:

$$\theta_0 = \tfrac{1}{4} \arg \sum_i e^{4i\varphi_i},$$

which maps all four Manhattan normals onto one angle, so the vote needs no association.

Measured: a frame is part of an estimator's output work in progress

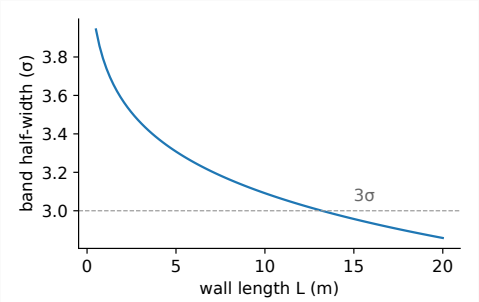
A fitted room with area 23.88 m^2 against 24.00 true, but aspect ratio 1.344 against 1.500. The fit was fine; the *frame* was wrong, because it was taken from the first scan. The cure: vote the gauge over all frames.

Which points belong to a wall? A band that is derived, not chosen

Clutter is uniform over the area A. A wall concentrates its points in a strip L long and about σ wide:

$$\frac{p_{\text{wall}}}{p_{\text{clutter}}} = \frac{A}{L\sqrt{2\pi}\,\sigma}\, e^{-r^2/2\sigma^2}$$

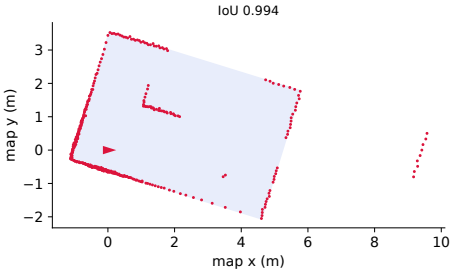
- Inlier iff ratio > 1: $r^2 < 2\sigma^2 \ln \frac{A}{L\sqrt{2\pi}\sigma}$, about 3σ.
- Each point brings ln(ratio) **nats**.
- The band **narrows as L grows**: a long wall is a more expensive explanation.



A tuned threshold gives the band. The derivation gives the band *and* the currency.

source: layout_acquisition.tex l.563–600

One scan, four walls



In the lab:

1. vote θ_0 from local normals;
2. initialise each class's d from a median;
3. refine each wall by a Laplace fit in which every point carries its **responsibility** $w = \rho/(1+\rho)$, entering with $\sigma/\sqrt{w}$;
4. intersect the walls in cyclic order.

The table, the bottle and the beams through the door are *explained away as clutter*. Nobody wrote an `if`.

The price of a shape

A polygon with more sides always fits better. What stops the estimator adding them?

Minimum description length. To transmit one edge, say its Manhattan direction and its offset at the sensor's resolution:

$$L_{\text{edge}} = \ln 4 + \ln \frac{2R}{c} = 1.386 + 5.298 = 7.29\,\text{nats} \qquad (R = 10\ \text{m},\ c = 8\ \text{cm}).$$

- An oblique edge names its angle instead of its class: a surcharge of 5.7 nats. Manhattan is *preferred*, not imposed.
- The old design had a hand-tuned 15 nats for a two-edge change. The derivation gives 14.6. It does not refine the constant; it explains it.
- The price depends on the instrument (R, c), not on the room. That is where it belongs.

source: layout_acquisition.tex l.702–738

But the table pays the price, too

- In the lab, the table's south face earns about **80 nats** as a candidate wall, against a price of 7.6. Face evidence alone would make the table part of the room.
- What tells a table face from a wall is **free space**: beams pass beside and beyond it and hit the real wall behind it.
- Hence two evidence channels, kept apart: face evidence, and a free/occupied grid. A single-number grid once marked 38 of 42 cells of a 13 cm wall as free.

Measured: the layout is a box cover of free space work in progress

Constructing the room from swept free space beat fitting a hull and repairing it on 8 of 9 test rooms (worst case 0.872 → 0.940 IoU). Free space must be charged both ways: area claimed but never swept, *and* area swept but not claimed.

source: layout_acquisition.tex l.329–415

Localisation is one more belief

The observation model: no rendering

For a scan $\{p_i\}$ in robot coordinates and a candidate pose $X = (R(\phi), t)$:

$$d_i(X) = \text{dist}\big(R(\phi)p_i + t,\ \text{polygon}\big), \qquad \mathcal{L}_{\text{obs}} = \frac{1}{2\sigma^2} \cdot \frac{1}{N} \sum_i w_i \,\text{Huber}(d_i)$$

- The residual *is* the prediction error: no expected scan is rendered. Its Jacobian is the gradient of the distance field.
- $w_i^{\text{range}} \propto r_i^{\alpha}$: far returns have more leverage on heading. $w_i^{\text{inc}} = |\hat{r}_i \cdot \hat{n}_i|^{\beta}$: Day 1's incidence lesson.
- Note the $\frac{1}{N}$. The term is a **mean**, so its precision does not grow with scan density. Keep that in mind until the end of the day.

source: room_concept.tex l.241–310

What the residual floor is made of

Park the robot. The mean wall distance of each new scan hovers near **87 mm**, and never reaches zero.

- The fluctuation is white, and follows $N^{-1/2}$: measured ratio 1.41 against a predicted $\sqrt{405/200} = 1.42$.
- Solved backwards, that gives a per-point scatter of ≈ 37 mm, which accounts for $0.798 \times 37 \approx 30$ mm of the floor.
- The other ≈ 57 mm is **map mismatch**: the room is not exactly the polygon.

Two-thirds of the floor is an error every return shares. More returns cannot remove it.

source: room_concept.tex l.385–425

Predict honestly: no constant process noise

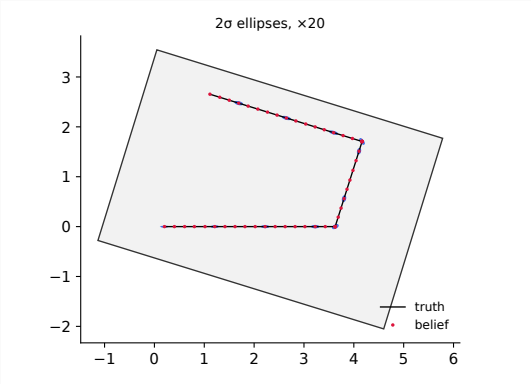
Between scans the pose is propagated by odometry:

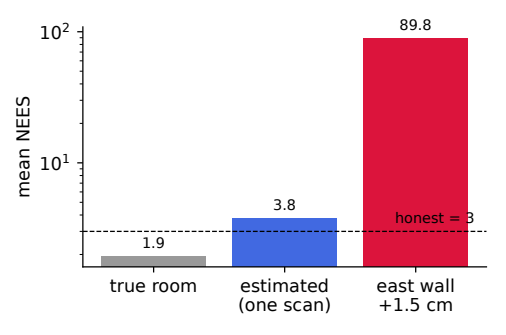
$$\Sigma_{t+1} = F\,\Sigma_t F^\top + G\,\mathrm{diag}(\sigma_v^2, \sigma_\omega^2)\,G^\top$$

- F, G: Jacobians of the motion with respect to the pose and to the measured (v, ω).
- The covariance grows because of what the encoders and gyro cannot tell, *not* by adding a constant Q every cycle.
- In the lab, dead reckoning with this model is calibrated: normalised errors have standard deviations of $1.13, 0.99, 1.07$, against 1 if honest.

A constant Q is a modelling decision dressed as a tuning knob. On Day 4 we will meet the pathology it causes in object beliefs.

Localising against the room you just built

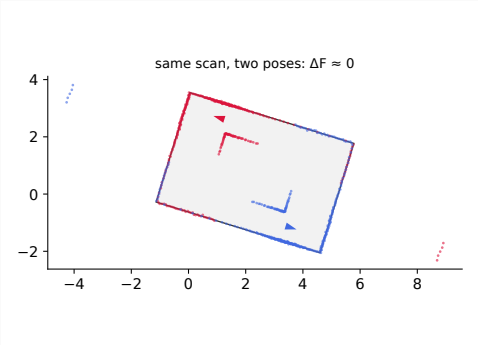




A wall **1.5 cm** out of place makes the localiser confidently wrong: NEES 90 where 3 is honest.

In the operating regime the polygon is a constant, so its error is shared by every scan, for ever.

Symmetry: where descent cannot help



- A rectangle looks the same after a half-turn. The same scan fits both poses with *identical* free energy.
- No downhill path joins them. They are **separate hypotheses**, so the agent must enumerate them and choose: a coarse search over the room's symmetries, then local refinement.
- This is one of the few places where a threshold is right: *a threshold is defensible exactly when the alternative cannot be expressed as a covariance.*
- What breaks the tie must be evidence the room does not contain: the motion prior, or an asymmetric landmark (the thesis uses the fridge).

source: room_concept.tex l.1867–1927

Shared error

The common-mode ceiling

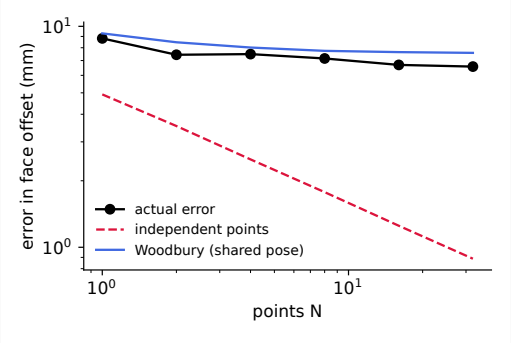
N residuals $r = r_0 + J_z\delta + \varepsilon$ that share one offset $\delta \sim \mathcal{N}(0, \Sigma_\delta)$ (a pose error), plus independent noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$. Marginalise δ:

$$\mathcal{I} = J_z^\top \left(\sigma^2 I + J_z \Sigma_\delta J_z^\top\right)^{-1} J_z \overset{\text{Woodbury}}{=} \left(\sigma^2 (J_z^\top J_z)^{-1} + \Sigma_\delta\right)^{-1} \xrightarrow[N\to\infty]{} \Sigma_\delta^{-1}$$

- A denser mask eventually stops helping. The ceiling is set by the *pose* error, not by the pixel count.
- The joint information on (object, offset) has rank 6, not 12: the data cannot tell a displaced object from a displaced robot. Only the ceiling is computable.

source: concept_agents.tex l.1047–1090

The ceiling, measured in the toy



The robot maps the table's face from a *dead-reckoned* pose, before it has a room to localise against: exactly the situation while a room is still being estimated.

- Treating points as independent claims under 1 mm at $N = 32$.
- The real error stalls near 7 mm.
- Woodbury predicts it to within about 10% at every N.

Averaging removes independent error. It does not remove shared error.

Case study: association dies by a common-mode latch

work in progress

One frame of a real recording, four walls still associating:

seg	χ^2	$\Delta\phi$	gate σ_ϕ
0	1.03	$-5.85°$	$6.78°$
1	1.02	$-5.76°$	$6.34°$
2	1.20	$-5.79°$	$6.01°$
3	1.06	$-5.77°$	$6.67°$

- The *same* $-5.8°$ on every wall is one **heading** error, not four wall errors.
- Each wall is gated independently against an error they all share. When the common error crosses $\approx 6°$, they all fail in the same frame.
- With nothing associated, nothing is left to correct the pose: 100% $\rightarrow$ 0%, permanently.

Parked runs never accumulate 6°, which is why "a parked run looks perfect and the first tour kills it".

source: post-thesis note, 2026-09-18

Case study: the pose must not lean on a room still being built

work in progress

While the room was being estimated, the pose was registered against the *provisional* layout. Layout errors dragged the pose; the dragged pose corrupted the layout.

pose error (median, vs ground truth)	run A	run B (20 min)	run C (wheel slip)
registered to the provisional layout	0.22 m	3.13 m	1.55 m
odometry alone	0.08 m	0.20 m	1.77 m
plain scan-to-map SLAM	0.010 m	0.013 m	0.031 m

- The fix: take the pose from plain SLAM, and build the layout *from* the aligned map.
- Live in simulation: pose median 6 mm; layout IoU 0.896 at 20 s and 0.956 at 60 s.
- Registering to the provisional layout was *worse than odometry alone*: a factor inherits the errors of whatever it is estimated through (Day 1).

source: post-thesis note, 2026-10-09

Lab

Lab 2: the room, the pose, and the ceiling

`labs/day2.ipynb`, about 75 minutes. No labels, no known pose.

A. **Acquire**: the 4θ vote; the derived band; four walls by responsibility-weighted Laplace fits; the polygon and its IoU. MDL: does the table pay the price?

B. **Localise**: the pose residual; drive the route; NEES against the true room, the estimated room, and a room with one wall 1.5 cm out.

C. **Symmetry**: F at the true pose and at its half-turn. What breaks the tie?

D. **Ceiling**: map the table face from a dead-reckoned pose; actual error vs independent vs Woodbury.

Tomorrow

Today the robot's odometry was honest. Tomorrow it is not: its wheels over-report by a few per cent and its gyro drifts. Nobody tells it so.

It will discover its own body from the corrections the room supplies, learn which motions reveal which parameters, and find that standing still is sometimes the most informative thing it can do.

Reading for Day 3

Thesis chapter *Self-Calibration of the Robot's Body*: "What the body is", "What no data can determine", "What each motion makes observable".